Module 3 Assignment 3

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Assignment Details

Purpose

The goal of this assignment is to assess your ability to perform and interpret linear regressions and to perform a statistical analysis for your research question.

Task

Write R code which produces the correct answers and correctly interpret the results of visualizations and statistical tests.

Criteria for Success

- Code is within the provided code chunks
- Code chunks run without errors
- Code produces the correct result
  - Code that produces the correct answer will receive full credit
  - Code attempts with logical direction will receive partial credit
- Written answers address the questions in sufficient detail

Due Date

Nov 6 at 10:45 am MST

Assignment Questions

In this assignment, we are going to continue using the hair grass data set from class.

At the end of the assignment, you will perform the correct statistical analysis for your own research question.

NOTE! Because you are choosing which variables to use in both sections of the assignment, there is no Answer Key provided.

Section 1

Set-Up

As always, we must get organized before we can do anything!

First, load the `tidyverse` and read in the hair grass data set.

```
# Load the tidyverse
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.2
## v ggplot2   4.0.0     v tibble    3.3.0
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.1.0
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
# Read in the dataset
hairgrass <- read_csv("../data/hairgrass_data.csv")
```

```
## Rows: 240 Columns: 8
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## dbl (8): hairgrass_density, soil_nitrogen, soil_phosphorus, penguin_density,...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Linear Regression with Hairgrass Data

Take a look at the hairgrass data and choose one of the environmental variables we think might influence hairgrass density that we have not used in class (AKA choose one *other than* nitrogen content or soil pH).

1. Calculate the mean and standard deviation of your chosen environmental variable.

```r
# soil phosphorus
hairgrass |>
  summarise(mean_phosphorus = mean(soil_phosphorus),
            sd_phosphorus = sd(soil_phosphorus))
```

```
## # A tibble: 1 x 2
##   mean_phosphorus sd_phosphorus
##             <dbl>         <dbl>
## 1            4.98          1.51
```

```r
# penguin density
hairgrass |>
  summarise(mean_penguins = mean(penguin_density),
            sd_penguins = sd(penguin_density))
```

```
## # A tibble: 1 x 2
##   mean_penguins sd_penguins
##           <dbl>       <dbl>
## 1          15.4        3.99
```

```r
# avg summer temp
hairgrass |>
  summarise(mean_tempC = mean(avg_summer_tempC),
            sd_tempC = sd(avg_summer_tempC))
```

```
## # A tibble: 1 x 2
##   mean_tempC sd_tempC
##        <dbl>    <dbl>
## 1       1.50    0.103
```

```r
# avg windspeed
hairgrass |>
  summarise(mean_windspeed = mean(avg_windspeed),
            sd_windspeed = sd(avg_windspeed))
```

```
## # A tibble: 1 x 2
##   mean_windspeed sd_windspeed
##            <dbl>        <dbl>
## 1           8.17         1.64
```

```r
# percent rock
hairgrass |>
  summarise(mean_rock = mean(percent_rock),
            sd_rock = sd(percent_rock))
```

```
## # A tibble: 1 x 2
##   mean_rock sd_rock
##       <dbl>   <dbl>
## 1      30.0    1.99
```

Now, we are going to explore the relationship between your chosen variable and hairgrass densities.

2. Which variable is the independent variable? Which is the dependent?

   Independent: Whatever environmental variable they chose above

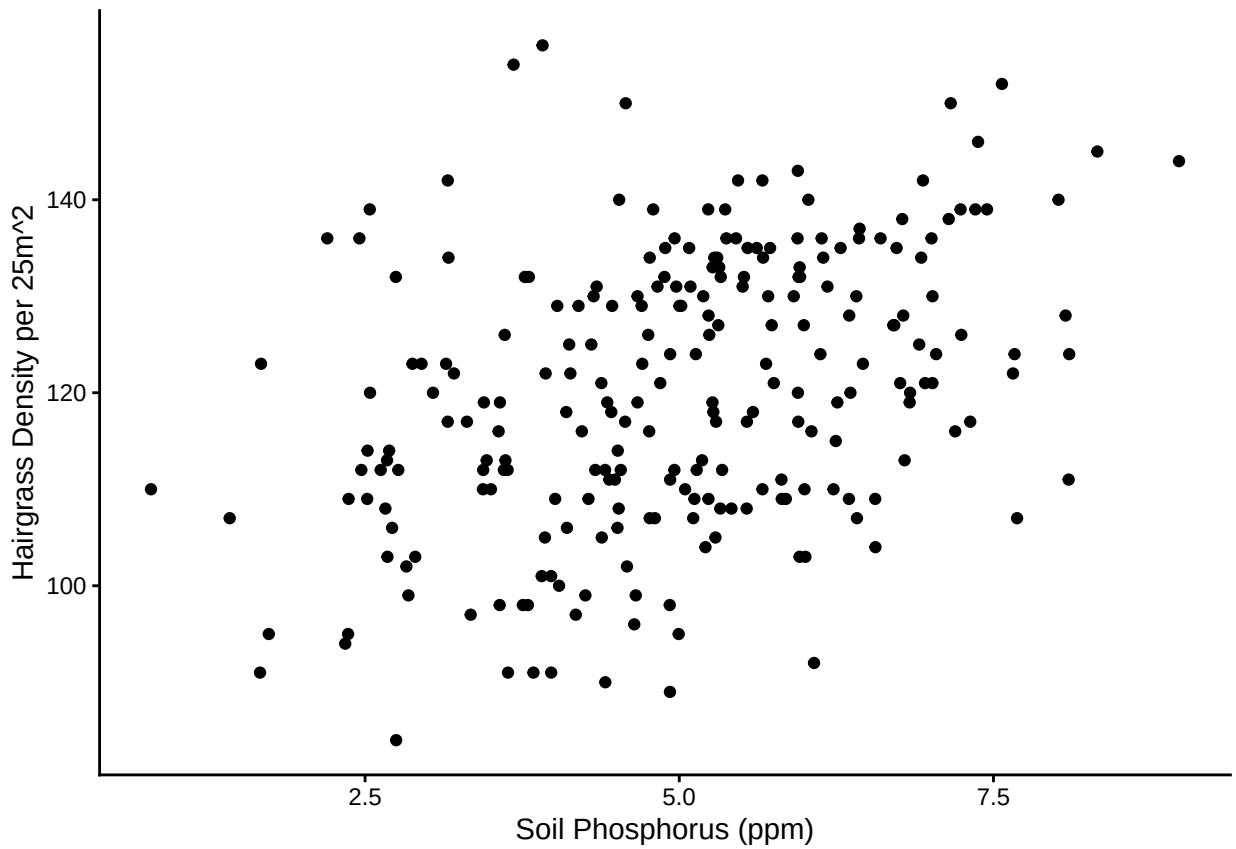
   Dependent: Hairgrass density

3. Create a scatter plot of hair grass density and the environmental variable (do not add a line of best fit yet).

   Be sure to make the labels easier to understand and add a theme. Also make sure that the dependent variable is on the y-axis.

```r
# Plot should have environmental variable on the x-axis and hairgrass density on the y-axis

# Example
ggplot(hairgrass, aes(x = soil_phosphorus, y = hairgrass_density)) +
  geom_point() +
  labs(x = "Soil Phosphorus (ppm)", y = "Hairgrass Density per 25m^2") +
  theme_classic()
```



4. Write 1-2 sentences interpreting the plot above. Is this a positive relationship, negative relationship or no relationship at all? Based on your prediction, do you think the correlation coefficient will be positive, negative, or zero?

   Answer:

   - soil phosphorus: positive, positive
   - penguin density: either positive or no relationship; positive or zero (unclear from plot alone)
   - avg summer temp: none, zero
   - avg windspeed: none, zero
   - percent rock: unclear, but generally appears to be none, zero

5. Calculate the correlation coefficient, `r`.

```
cor(x = hairgrass$soil_phosphorus, y = hairgrass$hairgrass_density)
```

```
## [1] 0.3847593
```

```
cor(x = hairgrass$penguin_density, y = hairgrass$hairgrass_density)
```

```
## [1] 0.2291056
```

```
cor(x = hairgrass$avg_summer_tempC, y = hairgrass$hairgrass_density)
```

```
## [1] 0.05228267
```

```
cor(x = hairgrass$avg_windspeed, y = hairgrass$hairgrass_density)
```

```
## [1] -0.05014076
```

```r
cor(x = hairgrass$percent_rock, y = hairgrass$hairgrass_density)
```

```
## [1] -0.01690271
```

6. Calculate the `r^2` value. Write a one sentence interpretation of what the `r^2` value means in the context of these two variables. (2 points)

```r
# saving output as r and then squaring it here is acceptable
# it does not matter whether students multiple it by 100 or not
cor(x = hairgrass$soil_phosphorus, y = hairgrass$hairgrass_density)^2 * 100
```

```
## [1] 14.80397
```

```r
cor(x = hairgrass$penguin_density, y = hairgrass$hairgrass_density)^2 * 100
```

```
## [1] 5.248938
```

```r
cor(x = hairgrass$avg_summer_tempC, y = hairgrass$hairgrass_density)^2 * 100
```

```
## [1] 0.2733478
```

```r
cor(x = hairgrass$avg_windspeed, y = hairgrass$hairgrass_density)^2 * 100
```

```
## [1] 0.2514095
```

```r
cor(x = hairgrass$percent_rock, y = hairgrass$hairgrass_density)^2 * 100
```

```
## [1] 0.02857015
```

Interpretation: According to the R^2 value, [amount] of variation in the hairgrass densities is explained by [independent variable]

7. What are the null and alternative hypothesis regarding the relationship between these two variables?

   Null: There is no relationship between hairgrass densities and [independent variable]

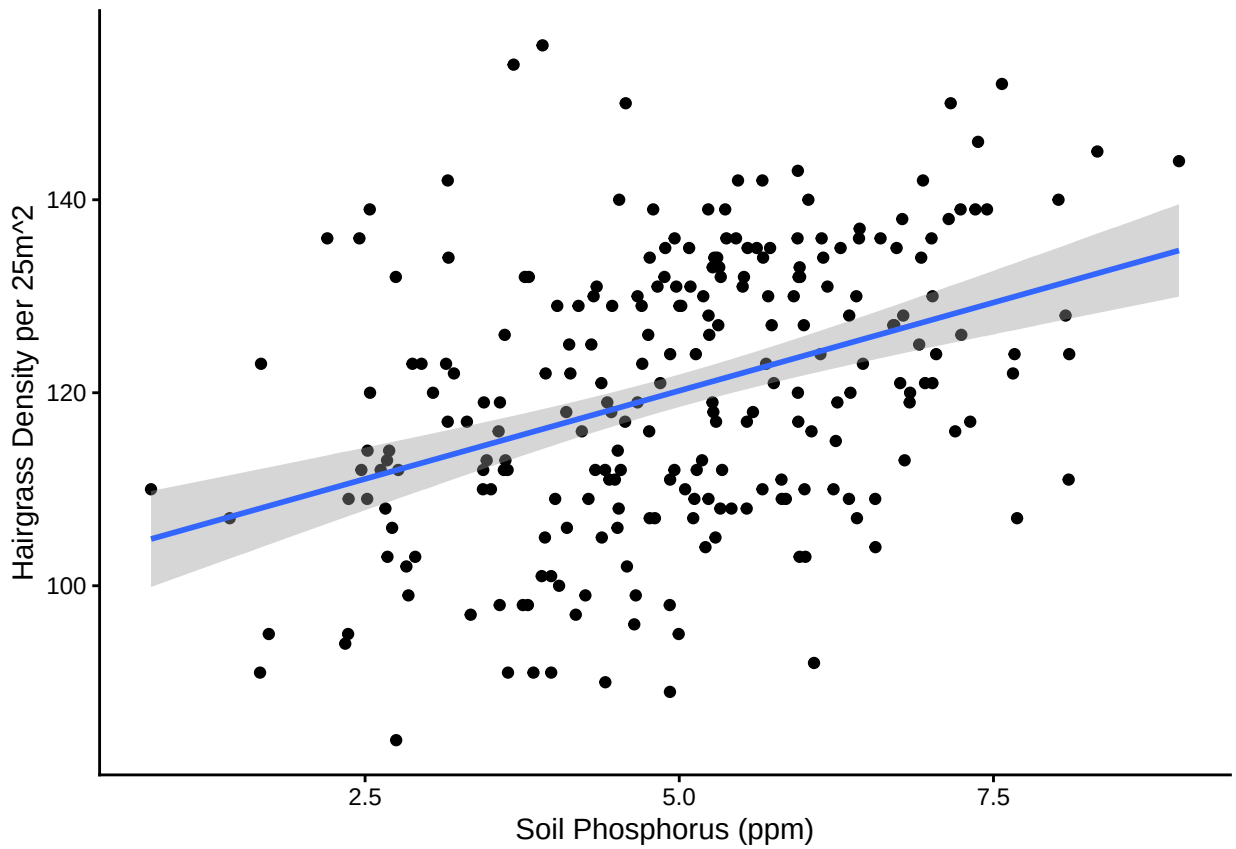
   Alternative: There is a relationship between hairgrass densities and [independent variable]

8. Create a scatter plot that includes the line of best fit. Again, make the labels clearer, add a theme, and make sure the correct variables are on the correct axes.

```r
# Plot should have environmental variable on the x-axis and hairgrass density on the y-axis

# Example
ggplot(hairgrass, aes(x = soil_phosphorus, y = hairgrass_density)) +
  geom_point() +
  geom_smooth(method = "lm") + # method = "lm" required, se = FALSE optional
  labs(x = "Soil Phosphorus (ppm)", y = "Hairgrass Density per 25m^2") +
  theme_classic()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



9. Run a linear regression model and print out the summary.

```r
# Take off -0.5 if independent and dependent variables are switched
# But for future questions, do not deduct points if they use the correct values

# soil phosphorus
model_phosphorus <- lm(hairgrass_density ~ soil_phosphorus, data = hairgrass)
summary(model_phosphorus)
```

```
##
## Call:
## lm(formula = hairgrass_density ~ soil_phosphorus, data = hairgrass)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -32.128  -9.889   0.371  10.120  39.768
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)     101.9321     2.9555  34.489  < 2e-16 ***
## soil_phosphorus   3.6552     0.5684   6.431 6.89e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.25 on 238 degrees of freedom
## Multiple R-squared:  0.148,  Adjusted R-squared:  0.1445
## F-statistic: 41.36 on 1 and 238 DF,  p-value: 6.887e-10
```

```r
# penguin density
model_penguins <- lm(hairgrass_density ~ penguin_density, data = hairgrass)
summary(model_penguins)
```

```
##
## Call:
## lm(formula = hairgrass_density ~ penguin_density, data = hairgrass)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -33.293  -9.496   0.473  10.798  33.061
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)     107.4176     3.6141  29.722  < 2e-16 ***
## penguin_density   0.8229     0.2266   3.631 0.000346 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.98 on 238 degrees of freedom
## Multiple R-squared:  0.05249,    Adjusted R-squared:  0.04851
## F-statistic: 13.18 on 1 and 238 DF,  p-value: 0.0003456
```

```r
# avg summer temp
model_summer_temp <- lm(hairgrass_density ~ avg_summer_tempC, data = hairgrass)
summary(model_summer_temp)
```

```
##
## Call:
## lm(formula = hairgrass_density ~ avg_summer_tempC, data = hairgrass)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -37.393 -10.441   0.254  11.444  35.720
##
## Coefficients:
##                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)       109.274     13.467   8.114 2.61e-14 ***
## avg_summer_tempC    7.258      8.986   0.808     0.42
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 14.34 on 238 degrees of freedom
## Multiple R-squared:  0.002733,   Adjusted R-squared:  -0.001457
## F-statistic: 0.6524 on 1 and 238 DF,  p-value: 0.4201
```

```r
# avg windspeed
model_windspeed <- lm(hairgrass_density ~ avg_windspeed, data = hairgrass)
summary(model_windspeed)
```

```
##
## Call:
## lm(formula = hairgrass_density ~ avg_windspeed, data = hairgrass)
##
## Residuals:
```

```
##     Min      1Q  Median      3Q     Max
## -35.732 -10.174  -0.276  11.431  35.355
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)   123.6989     4.7063  26.284   <2e-16 ***
## avg_windspeed  -0.4372     0.5645  -0.775    0.439
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 14.34 on 238 degrees of freedom
## Multiple R-squared:  0.002514,   Adjusted R-squared:  -0.001677
## F-statistic: 0.5999 on 1 and 238 DF,  p-value: 0.4394
```

```r
# percent rock
model_rock <- lm(hairgrass_density ~ percent_rock, data = hairgrass)
summary(model_rock)
```

```
##
## Call:
## lm(formula = hairgrass_density ~ percent_rock, data = hairgrass)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -36.033 -10.436   0.114  11.741  36.026
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  123.7721    14.0149   8.831  2.3e-16 ***
## percent_rock  -0.1215     0.4658  -0.261    0.794
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 14.36 on 238 degrees of freedom
## Multiple R-squared:  0.0002857,  Adjusted R-squared:  -0.003915
## F-statistic: 0.06802 on 1 and 238 DF,  p-value: 0.7945
```

10. Write out the equation for the line of best fit using *numeric values* from the results above and the *variable names*.

    Answer:

11. Interpret the model summary. What is the p-value for our variable of interest? Do we reject or fail to reject the null hypothesis regarding the relationship between these two variables? What can we conclude, then, about building a road? (2 points)

    Answer:

Section 2

Running A Statistical Model for Your Research Question

In this module, we have covered 3 different statistical analyses: t-tests, ANOVAs, and linear regression. It is now time to decide how to apply these statistical analyses to your own research question!

For this part of the assignment, referring back to **Module 2 Assignment 3** will be very helpful!

Set-Up: Research Question and Data

First, write your research question below (this really helps us when grading!).

Research question:

Now that you've written out your research question above, read in your chosen dataset for your final project.

Conceptual

12. Take a look back at Module 2, Assignment 3; this is the assignment where you explored your choices for data sets to use for the final assignment. Between questions 2 and 3, there is a lot of very important information about how you should treat certain variables in each dataset.

    Assisted by that information, determine whether each variable in your research question is numeric or categorical.

Answer:

13. State which variable is dependent and which one is independent.

Answer:

14. Based on your answers to Questions 12 and 13 above and other information about your chosen dataset, determine which statistical analysis you should perform to answer your question. In 1-2 sentence, explain how you know.

Answer:

Analysis

15. Perform the appropriate statistical analysis below. If necessary, add an additional line of code to print out the results.

16. If necessary, run a pairwise comparison. If a pairwise comparison is unnecessary for your statistical analysis, explain why.

Answer:

```
# space for code to run a pairwise comparison if needed
```

17. Interpret the results of your statistical analysis (including the pairwise comparison, if relevant). In your answer, include the p-value and if the result of your analysis is significant. Do you reject or fail to reject your null hypothesis? What do these results mean in the context of your research question? (2 points)

Answer: